2020

cmOS Engineering LCE14 User Manual

cmOS Engineering

1. Contents

Thank you for using a cmOS Engineering LCE14 Stall Motor Driver. Please ensure you read the sections on setting the jumpers and power requirements as a minimum before installing the LCE14 in your layout

Functional Description

The LCE14 as supplied, provides 4 output connections and 8 input connections specifically designed to interface to stall turnout motors such as DCC Concepts Cobalt © and Circuitron© Tortoise.

The eight inputs are used as feedback from the motors to indicate actual state.

Physical Description

The LCE14 is a 50mm by 40mm Printed Circuit Board (PCB), with a 20 pin interface connector which plugs into either expansion port on a cmOS Engineering LCP1 LCC node; an eight way screw connector (P1A /P1B ... P4A/P4B) which connects to the motor contacts on the turnout motor; a ten way screw connector (S1A/S1B ... S4A/S4B) which connects to one of the two status switched on the turnout motor; and a two way screw connector (+12/G) to allow for an external 12 volt power supply if desired.

Mounting

The LCE14 has four mounting holes.   It is recommended that at least two of these be used. Small neoprene spacers MUST be used. DO NOT screw any PCB directly to the layout structure as mechanical stress damage may occur.

Power Requirements

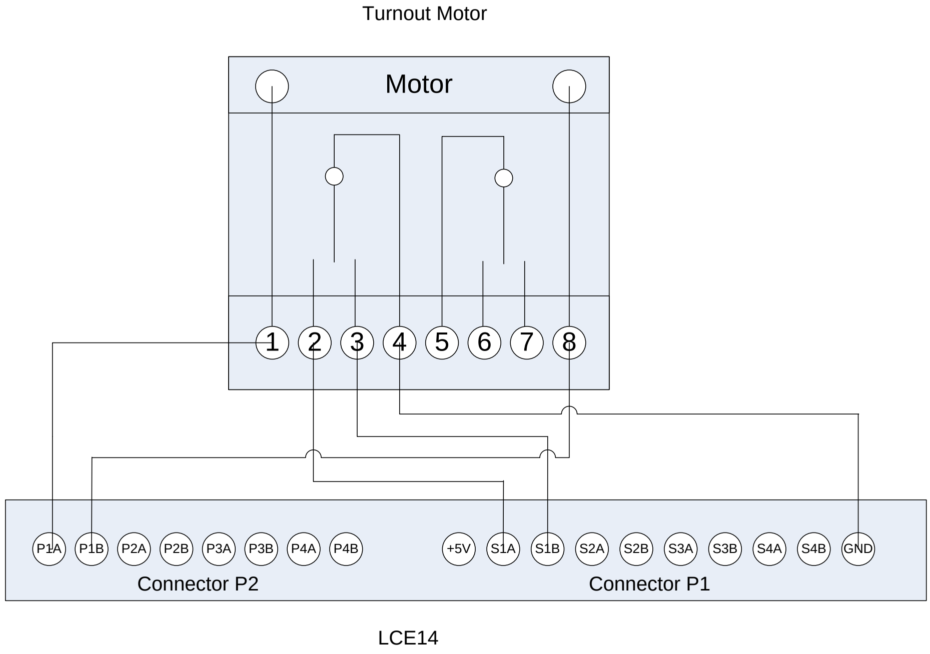
The basic power consumption of the LCE14 without any motors connected is less than 5mA. This power is supplied from the host LCP1.

Each turnout motor connected adds approximately 15ma. This is easily provided by the host LCP1 either from the LCC CAN bus or a local supply connected to the LCP1.

Power for the connected motors can also be supplied from a separate 12 volt supply connected to the LCE14 two way screw connector. The only jumper on the LCE14 selects between these two options and is shipped in the internal position.

Wiring

Each turnout motor is wired as below. The GND on P1 is common for all four turnout motors; all other connections will only have a single wire. The +5V on P1 is for courtesy only and in normal application have no use. Ensure the motor connections are wired between the PxA and PxB connections. Do not GROUND either side of the Turnout Motor.



Configuration

<u>Note:</u> Configuration involves writing to the configuration memory of the LCP1 host. This will cause a functional reset which may cause accessories to change state, possibly causing derailments etc. Do not configure when trains are running.

LCP1 i/o for the LCE14 is Bits 1..4 are outputs to drive the motor; bits 5..8 are inputs to provide status indication.

For the purpose of this section, the motor is assumed to be in the Normal (Unturned, Unthrown, Unreversed state) depending upon your railroad parlance, when PxA is connected to pin 1 of the x Tortoise and when the LCP1 configuration for the LCE14 is in the default state.

The LCP1 configuration for the LCE14 will default to the factory settings in the LCE14. With the default LCP1 settings, the motor will be in this Normal position when the power is applied. Due to the myriad of track patterns, these configurations can be changed to suit the turnout location and power on state.

If the turnout is required to be in the other state at power on, the connections to 1 and 8 on the motor can be reversed.

If the sense of the status indication is required to be in the other state, the connections to 2 and 3 on the motor can be reversed.

Alternatively, the logic on the output or input connections on the LCP can be reversed. It is suggested to change the wiring rather than change the LCP1 configuration but the choice is yours.

If the LCE14 is being used by an automation system, be aware of the timing of the change of status. The Tortoise motor takes 1.8 seconds approximately to change state. The mechanical timing of the switches appears to take .7 second to change state. The software in the microprocessor on the LCE14, waits to see both switches change state. This means the turnout will still travel for .7 of a second even though the sensors advise the turnout has changed.

cmOS Engineering products

cmOS Engineering Part Number	Description	Comments
LCP1	Primary LCC bus node	LCC node with 16 ports each capable of being producer or consumer. 2 sockets to allow any combination of LCExx cards being connected allowing up to 32 producers or consumers.
LCE08	8 input/output high current	Provides 8 channels individually programmable as producer or consumer. Consumer mode allows current of up to 500mA per channel - although total limit of 1 amp per board. Plugs directly into LCP1
LCE14	4 channel stall motor driver	Provides 4 channels of stall motor drive with status feedback to indicate actual turnout position. Each port can drive 2 parallel turnout motors. Plugs directly into LCP1
LCE24	4 channel block detector	Provide 4 channels of block detection. Plugs directly into LCP1. DCC systems only
LCE34	4 channel pulse point driver	Provides 4 channels of pulse motor drive with status feedback to indicate actual turnout position. Each port can drive 2 parallel turnout motors. Plugs directly into LCP1
LCEUSB	USB to CAN adaptor	Provides USB interface to computer system for configuration or enhanced layout automation.

LCEGA1*	Combinational Logic array	Provides virtual programmable logic gates for complex control without the need for a computer system. Connects to the LCC bus.
SMD2	Stand alone dual stall motor driver	Small standalone control for single wire control of stall motor turnouts
BD4	Stand alone quad block detector	Provides 4 independent channels of block detection. Open collector output for control loads of 100mA. DCC systems only.

* Product currently in design

Specifications

Mechanical		
	Dimensions	Width: 50mm Height: 60mm
	Mounting	4 x 3.5mm holes
	Weight	
Environmental		
	Operating	-40°C to 85°C; Humidity <90%
	Storage	-60 °C to 150 °C; Humidity <90%
Electrical		
	Absolute Maximums	Supply voltage 13.8 v Voltage on any input/output pin: Max +5.2V Min -0.2V DC current per input/output pin: 35mA total per device 300mA
	Internal Supply voltage	Min 7 Volts; Max 12.8 Volts Reverse voltage protected
	External Supply Voltage	13.8 Volts - Not Reverse voltage protected
	Connectors	1 x 2 Screw Terminal for connection of ecternal power supply to power the Turnout Motors 1 x 8 Screw Terminal for connection to Turnout Motor 1 x 10 Screw Terminal for input connection from Turnout Motor with +5V & GND 1 x 20 pin Interconnect to cmOS Engineering LCP1
	CAN termination	Passive bypassed split resistors (Jumper option)
Standards		Not Applicable

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In alphabetical order:

Digital Graphics P/L

References

cmOS Engineering: CMOSEng.com.au

JMRI

Users: https://groups.yahoo.com/neo/groups/jmriusers/conversations/messages

Project: http://jmri.sourceforge.net/